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To cite this article: Wan Li, Bindong Sun, Tinglin Zhang & Zhifan Zhang (2022) Polycentric spatial structure and its economic performance: evidence from meta-analysis, *Regional Studies*, 56:11, 1888-1902, DOI: [10.1080/00343404.2021.2012142](https://doi.org/10.1080/00343404.2021.2012142)

To link to this article: <https://doi.org/10.1080/00343404.2021.2012142>



Published online: 18 Jan 2022.



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Polycentric spatial structure and its economic performance: evidence from meta-analysis

Wan Li^a , Bindong Sun^b , Tinglin Zhang^c  and Zhifan Zhang^d 

ABSTRACT

Although there seems to be a widely shared expectation that polycentricity increases economic competitiveness, the empirical findings are inconsistent. As the first meta-analysis, this paper examines whether emerging empirical evidence provides an insight into the polycentricity–productivity effect. Based on 139 estimates, we find no compelling evidence to support that polycentricity is either beneficial or detrimental for economic development. Further meta-logit regressions show that the measurement of polycentricity and the control of agglomeration economies indeed affect the polycentricity–productivity effect. In contrast, measuring polycentricity from different perspectives and the potential endogeneity do not appear to significantly influence this effect.

KEYWORDS

spatial structure; polycentricity; economic performance; meta-analysis

JEL O1, O2, R1

HISTORY Received 18 September 2020; in revised form 7 November 2021

INTRODUCTION

Polycentricity has been a favoured spatial pattern since the issuing of the European Spatial Development Perspective (Commission of the European Communities (CEC), 1999), and the establishment of China's *Xiong'an* New Area in 2017 has made polycentric spatial structure a hot topic again. This constant interest is fuelled by the shared expectation that polycentricity is an effective tool to enhance economic competitiveness (CEC, 1999; Davoudi, 2003; Vandermotten et al., 2008; Waterhout et al., 2005; Wegener, 2013). However, the empirical evidence for the economic effects of spatial structure is quite inconclusive. Although some studies have found that polycentric spatial structure, or its monocentric counterpart, barely affects economic performance (Lee & Gordon, 2007, 2011; Rauhut et al., 2018; Veneri & Burgalassi, 2012), others have supported spatial structure affecting economic performance in conflicting directions. In other words, both polycentricity and monocentricity have been found to be

better spatial layouts in terms of economic performance (Li et al., 2019; Li & Liu, 2018; Meijers & Burger, 2010; Zhang et al., 2017).

Insightful attempts have been made to explain this inconclusiveness. Brezzi and Veneri (2015) suggested that the conceptual definition of polycentricity (i.e., morphology or function) and the choice of geographical scale (i.e., micro, meso or macro) could produce variations in empirical findings. A very recent study conducted by Wang et al. (2019) confirmed that the impacts of polycentricity on economic productivity are opposite in different territorial scales under the same analytic frame. Unfortunately, these efforts are far from sufficient to fully clarify the differing results, for example, how to explain the different results of Lee and Gordon (2007) and Meijers and Burger (2010), which shared the same geographical scale and employed the same conceptual definition of polycentricity.

The article aims to render more lucid the relationship between polycentricity and economic performance with

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This article has been corrected with minor changes. These changes do not impact the academic content of the article.

meta-analysis. This approach provides an objective way to systematically review the empirical studies, summarize the often-conflicting results, and examine the effect of different study characteristics on the empirical results (Stanley & Doucouliagos, 2012; Stanley & Jarrell, 2005). To the best of our knowledge, our work is the first meta-analysis of the increasing number of empirical studies of polycentric spatial structure and its economic performance. This study contributes to this field in at least two important ways. First, this paper helps to answer the fundamental question of the extent to which polycentricity matters for its economic performance. Due to the conflicting research findings in previous studies, there is unlikely to be a short and clear answer to this question. Here, we attempt to provide an objective and comprehensive summary and an evaluation of the polycentricity–productivity effects. Additionally, potential publication selection bias is filtered out to obtain an unbiased assessment. Second, this paper provides a quantitative analytic assessment of why broad variation exists in polycentricity–productivity effects. By collecting 139 estimates reported in 31 studies, we systematically examine three potential explanations for why polycentricity can promote economic development or, conversely, why polycentricity fails. Other characteristics of the primary studies, such as the form and period of data, sample location and size, and publication-specific features, are also considered.

OVERVIEW OF THE LITERATURE

Polycentricity and economic performance

It is widely believed that a polycentric spatial layout is conducive to better economic performance, and there are two theoretical justifications for this conviction. First, cities within polycentric regions are usually smaller in size and therefore less prone to agglomeration diseconomies such as congestion, high housing prices, environmental pollution and crime (Meijers & Burger, 2010). Second, cities within a polycentric region can borrow ‘size’ and/or ‘function’ from each other due to the generally effective network of linkages (Alonso, 1973; Meijers & Burger, 2017). However, in consideration of agglomeration economies, some scholars argue that the economic performance of monocentric spatial structures, which is the counterpart of polycentric spatial layout, should be better (Li et al., 2019; Veneri & Burgalassi, 2011). Specifically, a typical monocentric spatial structure tends to feature a high level of agglomeration economies, and its accompanying sharing, matching and learning effects can make monocentricity more efficient. In addition, while individual cities within a polycentric region can also generate agglomeration economies, some studies question that a collection of nearby cities cannot provide a substitute for the agglomeration economies of a single larger city (Meijers, 2005). Thus, it is difficult to verify theoretically that a polycentric spatial structure leads to higher economic performance. This not only makes this relationship more or

less an empirical question, but also one that needs to be explored with meta-analysis.

Empirical model for the polycentricity–productivity effect

The standard approach to estimating the polycentricity–productivity effect is based on an extended Cobb–Douglas production framework (Li et al., 2019; Meijers & Burger, 2010; Zhang et al., 2017). Specifically, the total output (Y) is produced as a function of capital (K), labour (L), human capital (H) and an efficiency parameter (A):

$$Y = AK^k L^l H^h, \quad (1)$$

where A represents a vector of exogenous factors on the output, which can reflect the influence of polycentricity. In other words, within this framework, polycentricity impacts economic output because it is a component of efficiency. By assuming constant returns to scale and calculating natural logarithms on both sides, a typical linear specification can be given as:

$$\ln(Y/L) = \beta_0 + \beta_1 \ln(Poly) + \sum \beta_j \ln(Z_j) + k \ln(K/L) + h \ln(H/L) + \varepsilon_i, \quad (2)$$

where $Poly$ is the focal independent variable; and Z_j is a set of control variables that can be correlated with $Poly$. The coefficient of interest is β_1 , which is the estimate of the effect of polycentricity on economic production. A significantly positive sign implies that polycentricity is beneficial for economic competitiveness, and vice versa. In addition to the Cobb–Douglas production function, other estimation approaches have been used. For example, Lee and Gordon (2007) adopted a supply-side urban economic growth model to analyse how polycentricity affects population and employment growth.

Measurement of polycentricity

One of the most critical research dimensions consists of the measures of the key independent variable, that is, polycentricity, which can serve as a vital source for inclusive findings of the polycentricity–productivity effect. Although polycentricity has long been criticized for its undifferentiated catchall characteristics, there have been two dominant and distinct approaches, morphological and functional, to understand it (Hall & Pain, 2006; Van Meeteren et al., 2016). As Burger and Meijers (2012, p. 1128) stated, ‘a balance in the size distribution of centers does not necessarily imply that there are functional linkages between the different centers, let alone an equal distribution of these linkages and the existence of multidirectional flow patterns’. Hence, using different approaches to measure polycentricity can naturally lead to variation across studies. Most previous studies, however, focused on measuring polycentricity from the perspective of morphology, rather than function, due to the poor availability of interaction data.

Furthermore, the measurement methods introduced to assess the degree of polycentricity – in both the

morphological and functional sense – may also lead to distinct findings. According to Bartosiewicz and Marcińczak (2020) and Derudder et al. (2020), different indicators may not point to a similar pattern of polycentricity. Drawing on the classification of these two articles, we distinguish four groups of measures of polycentricity: (1) proportion-based indicators, for example, the primacy index; (2) rank size-based indicators; (3) standard deviation-based indicators, for example, the ordinary polycentricity proposed by Green (2007); and (4) other indicators, for example, the number of centres. In addition, there has been a tradition of distinguishing between a fixed number of centres or all of them when measuring polycentricity (Meijers & Burger, 2010; Zhang & Derudder, 2019).

Another critical aspect of polycentricity is its scale dependence: a system that can be monocentric on one geographical scale can be polycentric when measured on another (Hall & Pain, 2006; Möck & Küpper, 2020; Taylor et al., 2008). This scale dependence of polycentricity is considered the most evident source for the inconsistent estimations. For example, Brezzi and Veneri (2015) showed that polycentric regions have lower gross domestic product (GDP) per capita among Organisation for Economic Co-operation and Development (OECD) metropolitan areas and territorial level 2 (TL2) regions but higher GDP per capita at the OECD national level. Considering China's two different urban scales, Wang et al. (2019) found that both monocentricity at the intraurban level and polycentricity at the interurban level are linked to higher levels of labour productivity. Additionally, other subtle treatments, such as the choice of centres (Möck & Küpper, 2020; Zhang & Derudder, 2019), reference data (Li et al., 2018) and indicators (Li et al., 2019; Zhang et al., 2017), influence the measurement of polycentricity.

Control variable for agglomeration economies

The inclusiveness of the polycentricity–productivity effect in empirical studies might also come from the different sets of control variables (Z_j), especially whether they control for the 'agglomeration economies' variable. According to Anas et al. (1998), Davoudi (2003) and Boussauw et al. (2018), the polycentric pattern can emerge when agglomeration economies are exceeded by agglomeration diseconomies. Nevertheless, this assumption does not mean that polycentricity does not appreciate agglomeration economies. In fact, polycentricity is widely regarded as a pattern of 'concentrated decentralization' (Davoudi, 2003; Kloosterman & Musterd, 2001). After all, economic activities still prefer to relocate to concentrated centres, taking advantage of agglomeration economies, rather than being sparsely distributed (Anas et al., 1998). Furthermore, agglomeration economies also serve as a critical mechanism in answering whether polycentricity performs better in terms of economic productivity. Li et al. (2019), for example, found that agglomeration dominates its cost in China's prefectural region; thus, monocentricity enjoys a higher level of labour productivity. In addition, in the econometric specification, both the density and size of the population or employment are used as proxy variables

for agglomeration economies (Combes & Gobillon, 2015).

Estimation strategies for endogeneity

Estimation strategies can also yield conflicting findings. Endogeneity, the major concern in the empirical study of polycentricity and its performance, can arise when the variable *Poly* is associated with the random component ε_i , and the primary source of endogeneity is the potential of reverse causation. Specifically, higher productivity affects the spatial pattern of cities by attracting more firms and workers to some specific locations and dispersing people and firms into smaller cities with low land and real estate prices. A common approach is to introduce appropriate instrumental variables and run the two-stage least squares regression. Moreover, two different sets of instruments have been used in previous studies.

First, consider the historical instruments, that is, the long-lagged values of polycentricity (Li et al., 2019; Meijers & Burger, 2010). In practice, historical instruments are usually found to be perfectly relevant but to have questionable exogeneity, which is especially true when the lag is not sufficiently long. Second, geological instruments, such as the average distances of subcentres to the main centre in each city (Zhang et al., 2017), the average value of the slope, and the standard deviation of the landform curvature (Wang & Gao, 2018), have been used. Geological variables are believed to be extremely exogenous but are criticized for not being sufficiently powerful to explain instrumented variables.

Overall, this section identifies three often-criticized specification differences: the measurement of polycentricity (morphology or function; proportion-based indicator, rank size-based indicator, or standard deviation-based indicator; micro, meso or macro), the presence or absence of the control variable of agglomeration economies, and whether or not the potential endogeneity of polycentricity is addressed. These three aspects not only cover all important aspects of an empirical analysis model (the measurement of key variables, the selection of control variables, and the adoption of methodologies), but are also recognized considerations from previous studies that may result in different findings. Nonetheless, other characteristics of the primary studies, such as the form and period of data, the size and location of the sample, and the publication-specific features, could also lead to variation in research.

STUDY IDENTIFICATION AND DATA CODING

Meta-analysis is now widely used to review and assess previously published or reported empirical studies on a given hypothesis or empirical effect (Stanley & Doucouliagos, 2012). The first challenge to any meta-analysis is the identification of relevant studies. We begin by reviewing the reference and citation list of a well-known empirical study of polycentricity and its economic performance (for example, Meijers & Burger, 2010). In addition, we use Web of Science, Google Scholar and Cnki to detect

related papers with keywords such as ‘polycentric*’, ‘mono-centric*’ or ‘spatial structure’ combined with ‘economic performance’, ‘economic growth’ or ‘growth’. Eventually, we arrived at a sample of 139 observations from 31 different studies (Table 1).

The following should be noted.

- Studies written in either English or Chinese are included. As a small and emerging research topic, we would like to include as many primary empirical studies as possible to ensure the effectiveness and robustness of follow-up analysis. Moreover, the notion of polycentricity is gaining wide currency in China today (Cheng & Shaw, 2018), and an increasing number of high-quality studies are being published in Chinese academic journals (Liu et al., 2017; Zhang & Sun, 2017). Additionally, this difference can be modelled in the meta-regression analysis by including a dummy variable or setting an all-English subgroup.
- The term ‘economic performance’ is used in its general sense. We include papers with explanatory variables such as ‘labor productivity’ (Meijers & Burger, 2010), ‘GDP per capita’ (Li & Liu, 2018), ‘TFP’ (total factor productivity) (Ouweland, 2018), ‘population or employment growth’ (Lee & Gordon, 2007) and ‘the

number of retail stores or universities or pharmacies or hospital beds’ (Rauhut et al., 2018). Similarly, increasing the pool of primary empirical studies is the chief consideration.

- Multiple sampling is employed: taking multiple estimates from a single study is favourable for the power of the subsequent statistical tests and helps to produce more accurate estimates.
- Lee and Gordon (2011) is excluded because they reported only the interaction term of polycentricity.
- Studies of urban concentration and its economic growth are also not considered in our analysis. The key variable ‘urban concentration’ in studies such as by Henderson (2003) is a similar measure to polycentricity, but it usually takes both the linear and squared terms to capture possible nonlinear effects.

As having a focused theme is one of the keys to designing a high-quality meta-analysis, we use CiteSpace, a tool for visualizing progressive knowledge domains, to conduct a thematic analysis (Chen, 2004). Based on the keywords of the selected 31 studies (Table 1),¹ we obtain a keyword network of 76 nodes and 199 links (Figure 1). We find each keyword node is tied to the identical network. This suggests that the 31 articles we selected share a focused topic. Furthermore, according to the keywords with a higher centrality, polycentricity (0.82) and economic performance (0.82) truly dominate the network.

After selecting the reference studies, how to code the effect size is of pivotal importance in the next step. In this paper, we extract three different types of effect sizes to signify the relationship between polycentricity and economic development: elasticity, *t*-statistics and categorical result. As the most widely used measure of empirical economic effect, elasticity is only available when the original econometric model is estimated in double-log form. In our sample, we can collect only 98 elasticities from 20 studies. Such a considerable reduction in the sample is the main reason we resort to other types of effect sizes. *T*-statistics also have been commonly used to identify and correct for publication bias. However, the interpretation of *t*-statistics is limited in terms of economic meaning because it is just a statistical measure rather than an economic measure. Categorical result is the last effect size used in this paper to indicate the relationship between dependent and independent variables. Unlike previous continuous effects, this type of effect size focuses on whether a certain type of result is observed. In this way, the effect size of categorical result can include as many studies as possible.

DOES POLYCENTRICITY MATTER TO ECONOMIC PERFORMANCE?

Vote count

Vote count is a method of simply counting and comparing the number of estimates or studies that are significantly positive and those that are negative or insignificant. This approach provides a rough summary of the distribution of findings. Table 2 shows the results of vote counting

Table 1. Studies coded and included in the meta-analysis.

Brezzi and Veneri (2015)
Cai (2014)
Chen (2016)
Hou and Sun (2016)
Hua (2017)
Hua and Sun (2015)
Kwon and Seo (2018)
Lee and Gordon (2007, 2011)
Li and Liu (2018)
Li et al. (2018, 2019)
Liu et al. (2017)
Meijers (2008, 2013)
Meijers and Burger (2010)
Miao et al. (2016)
Ouweland (2018)
Rauhut et al. (2018)
Sun and Li (2016)
Sun et al. (2015)
Veneri and Burgalassi (2011, 2012)
Wang (2016)
Wang and Gao (2018)
Wang et al. (2019)
Yang et al. (2011)
Zhang (2012)
Zhang et al. (2017)
Zhang and Sun (2017)
Zhang and Yi (2012)

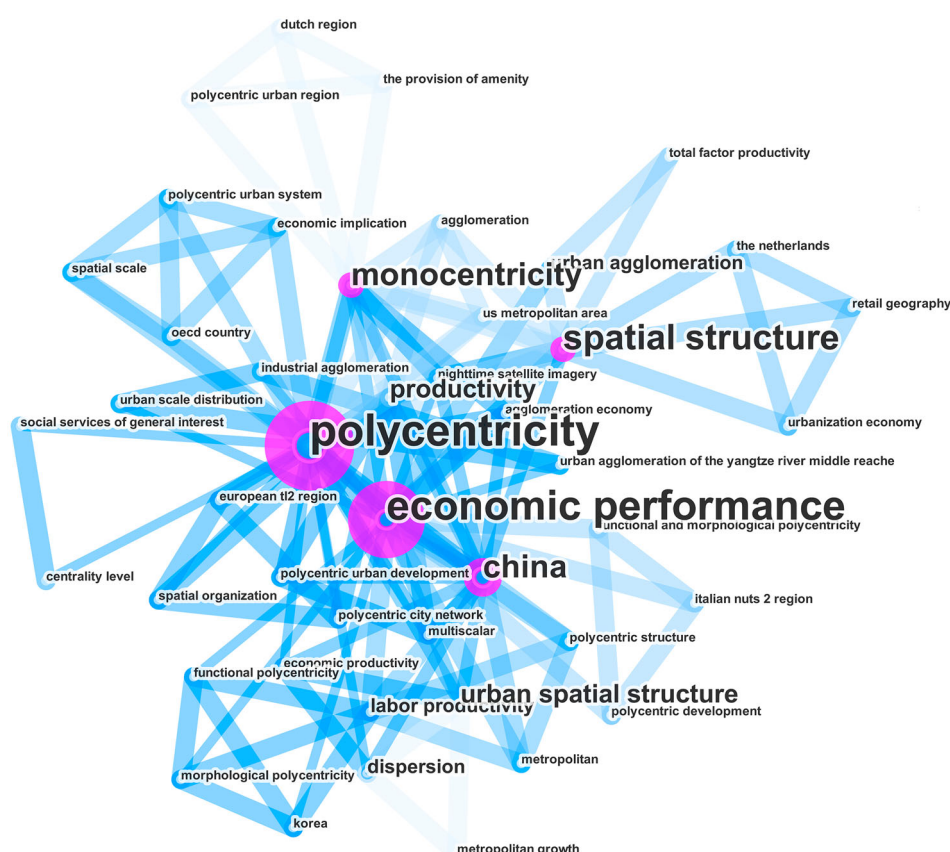


Figure 1. Visualization of the keyword networks.

Note: Each node represents one keyword; the thickness of the node is proportional to the frequency of the keyword; wide circles of the nodes indicate that the centrality of the keyword is > 1 , and shades of grey from light to dark indicate years from 2007 to 2019. The line between the nodes represents the co-occurrence of the keyword; the thickness of the line is proportional to the co-occurrence of the keyword; and the colour of the line represents the year in which the two keywords first appeared together. The detailed settings for CiteSpace are time slice: one year, node type: keyword and selection criteria: g -index ($k = 25$).

Table 2. Vote count of the polycentricity–productivity effect based on the three main specification differences.

		Significantly negative		Insignificant		Significantly positive	
		Estimates	Studies	Estimates	Studies	Estimates	Studies
Measurement of polycentricity	<i>Morphology</i>	50	13	28	12	43	13
	<i>Function</i>	14	3	3	2	1	1
	<i>Rank size-based measures</i>	27	11	17	10	22	9
	<i>Proportion-based measures</i>	27	4	6	3	5	2
	<i>SD-based measures</i>	9	4	6	2	16	5
	<i>Micro</i>	12	4	0	0	19	5
	<i>Meso</i>	37	9	15	6	9	6
	<i>Macro</i>	15	6	16	7	16	4
Control variable	<i>With_agg</i>	55	14	19	9	38	12
	<i>Without_agg</i>	9	2	12	4	6	2
Methodology	<i>Treated_endo</i>	26	6	8	6	16	5
	<i>Untreated_endo</i>	38	15	23	10	28	12
Studies written in English		43	9	21	8	16	5
Total studies		64	16	31	13	44	14

Note: Calculated based on 139 estimates from 31 studies; the number of studies is multi-counted if the study reported mixed results.

with respect to the polycentricity–productivity effect based on the three main specification differences.

Overall, while there is a weak tendency to support the negative effects of polycentricity, there is no definitive agreement as to whether polycentricity is an effective tool for promoting economic growth. According to Table 2, the number of estimates in the ‘significantly negative’ category is higher both in the total studies and studies written in English. However, a relatively large number of significant negative estimates may occur when some negative-effect studies reported robust results beyond the average number. This is confirmed by the only slightly higher number of studies in the ‘significantly negative’ category. Regarding the specific specification differences, 78% of estimates with functional polycentricity, 71% of estimates with proportion-based measures, and 61% of the estimates made at the meso-level found significantly negative results; that is, functional, proportion-based, and meso-level polycentricity tends to depress economic growth.

Funnel plot

A funnel plot is a scatter diagram of all empirical estimates versus their precision (i.e., the inverse of the estimates’

standard errors, $1/SE$) and is the most common graph used in meta-analysis (Stanley & Doucouliagos, 2012). The primary use of funnel plots is to illustrate publication bias in the literature, but they can also show, rather vividly, the wide variation in reported empirical results. Figure 2 presents a funnel graph of polycentricity elasticity for economic performance.

As shown in Figure 2, there is a large cluster of observations around zero, and most of the polycentricity elasticities for economic performance are in the range of -1 to 1 . This suggests a near-zero association between polycentricity and economic performance. Furthermore, while the average economic elasticity of polycentricity based on 98 initial estimates is -0.065 , the estimate at the top of the funnel, which is regarded as the most accurate or as precise elasticity, is -0.042 . This result implies a weak tendency for poorer economic performance of polycentricity. All these outcomes are in line with the vote counting findings; that is, while a slight tendency to support the negative effects of polycentricity exists, there are no strong signs that polycentricity is an effective tool to promote or inhibit economic development. Moreover, regarding the three main specification differences,

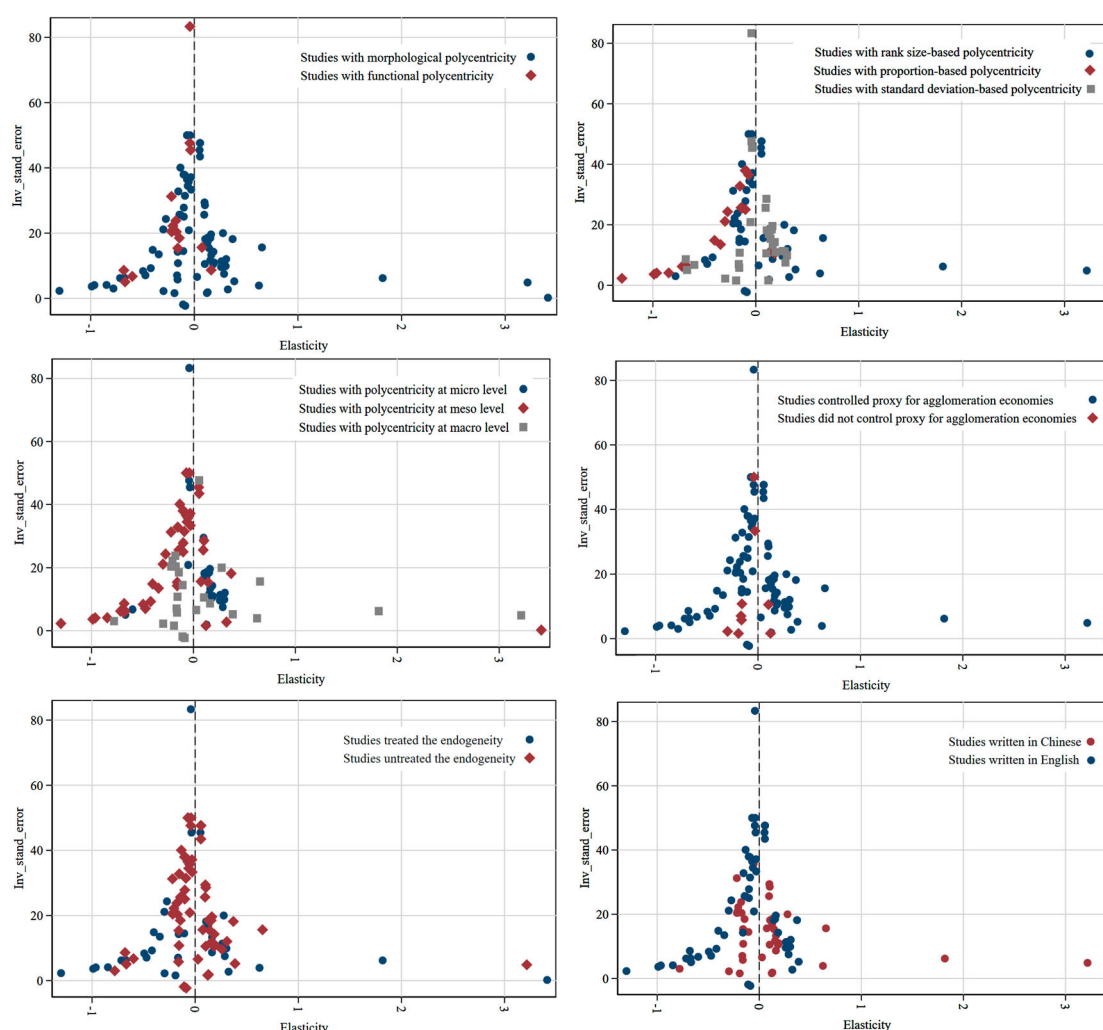


Figure 2. Funnel graphs of polycentricity elasticity for economic performance.

Note: Only studies with log–log are included in the funnel plot (98 estimations from 20 studies).

functional, proportion-based, and meso-level polycentricity and studies written in English presents a more noticeable negative tendency, which is also consistent with the observations made in vote counting. In addition, Figure 2 provides a rough approximation of the expected inverted funnel shape, which serves as a positive sign of the absence of publication bias, especially when outliers with elasticities > 1 are excluded.

Precision-effect test

While fairly straightforward, conclusions derived from vote counts and funnel graphs cannot be statistically validated. Here, we turn to a meta-regression model to assess the 'genuine' effect beyond publication selection bias and to reidentify publication bias. A more formal test based on t -statistics is conducted:

$$T_{ij} = \beta_1 + \beta_0(1/SE_{ij}) + v_{ij}, \quad (4)$$

where T_{ij} is the t -value of the estimated polycentricity–productivity elasticity from estimate i of study j ; and SE_{ij} is the corresponding standard deviation. Publication bias or a 'genuine' empirical effect exists if the constant β_1 or slope β_0 are significantly different from zero, respectively. Table 4 reports the test results for both the total sample and for subgroups by the three main specification differences.

According to the insignificant slope coefficient (β_0) in the first column of Table 3, there is no general genuine empirical effect after correcting for the potential distortion, irrespective of whether the outliers are considered. In other words, we find no statistical evidence of the polycentricity–productivity effect. Moreover, the corresponding intercept (β_1) is statistically non-significant, suggesting an absence of publication bias in the polycentricity–productivity studies. This result confirms our visual impression of the funnel graphs and is a positive sign that the expectation of polycentricity did not appear to influence the integrity of academic studies in this area.

For the three main specification differences, the regression results for most subgroups (columns 3–12) are generally consistent with the total and all-English samples (columns 1 and 2). Notably, the polycentricity–productivity effects at the micro-level and in the subgroup without treating endogeneity are slightly skewed toward negative values. Conversely, the weak negative tendency of the functional, proportion-based and meso-level polycentricity shown in the vote count and funnel graph fails the precision-effect test. Additionally, there is a possibility of publication bias in the previous empirical studies at the micro- and meso-levels. Notably, due to the limited sample size, the analysis of the three main specification differences requires further validation.

Overall, there is no compelling evidence that polycentricity is either beneficial or detrimental for economic development. Notably, this finding is a post-offsetting outcome between the positive and negative effects of polycentric structure on the economy, and the influence of

polycentric structure on economic development in specific cases cannot be negated.

WHY A WIDE VARIATION EXISTS IN POLYCENTRICITY–PRODUCTIVITY EFFECTS

Meta-logit regression

Meta-regression analysis, a multivariate empirical investigation of the existing pool of studies on a given empirical effect, can be used to quantify the impact of different study characteristics on research findings. Using the categorical result as the dependent variable, we group the polycentricity–productivity estimates into significantly negative, insignificant, and significantly positive and label them 1, 2 and 3, respectively. Our basic meta regression equation takes the following multinomial logit form:

$$\text{mlogit}(\text{Poly_result}) = F(\text{TCCSC}, \text{OSIC}, \text{PRC}), \quad (3)$$

The first group of explanatory variables primarily consists of the three commonly criticized specification characteristics (TCCSC):

- We introduce a dummy variable (*Meas_morph*) taking a value of 1 if polycentricity is quantified from the perspective of morphology.
- We classify the measurement methods of polycentricity into four groups (*Dum_method*): proportion-based indicator, rank size-based indicator, standard deviation-based indicator or others. A dummy variable (*Num_b_all*) is coded to capture whether the measurement is based on all centres or a fixed number of them.
- We differentiate the scale into three categories (*Dum_scale*): micro, meso and macro.

In light of the pool of primary empirical studies, we take polycentricity measured in Italian NUTS-3, Chinese prefect proper and Korean city as the micro-scale; polycentricity measured in the European TL2, Italian NUTS-2, Dutch Wet Gemeenschappelijke Regelingen (WGR) region, US metropolitan area, Chinese prefectural region and Korean functional urban region (FUR) as the meso-scale; and polycentricity measured in European countries, European TL1, Chinese provincial region, Chinese urban agglomeration region and Chinese neighbouring prefectural city region as the macro-scale. Considering the high relevance of scale and geography, we bound the three scale with specific geographical context (Asia, Europe and North America).

- A dummy variable (*Control_agg*) is coded for whether primary studies control for population or density.
- We mark the estimate with a dummy variable (*Endo_treated*) if it overcomes the endogeneity problem.

The next set of explanatory variables involves other study-invariant characteristics (*OSIC*).

Table 3. Precision-effect test.

$Y = T$	(1) All studies	(2) English	(3) Morphology	(4) Rank size	(5) Proportion	(6) SD
<i>Inv_stand_error</i> (β_0)	−0.043 (−1.59)	−0.010 (−0.40)	−0.041 (−1.32)	−0.069 (−1.34)	−0.025 (−0.51)	−0.027 (−0.65)
<i>Intercept</i> (β_1)	0.203 (0.16)	−1.292 (−0.95)	0.635 (0.44)	1.311 (0.65)	−3.120 (−2.19)	0.925 (0.66)
Number of estimates	98	60	82	48	20	29
Adjusted R^2	0.019	−0.014	0.011	0.039	−0.034	−0.005
F	2.513	0.157	1.749	1.802	0.262	0.425
$Y = T$	(7) Micro	(8) Meso	(9) Macro	(10) With_agg	(11) Treated_endo	(12) Untreated_endo
<i>Inv_stand_error</i> (β_0)	−0.061* (−2.05)	0.012 (0.53)	−0.030 (−0.34)	−0.045 (−1.43)	−0.016 (−0.57)	−0.076* (−1.93)
<i>Intercept</i> (β_1)	2.377* (2.40)	−2.741*** (−4.22)	1.483 (0.64)	0.283 (0.20)	−0.721 (−0.48)	1.275 (1.00)
Number of estimates	25	48	25	89	36	62
Adjusted R^2	0.161	−0.017	−0.037	0.018	−0.025	0.063
F	4.206	0.276	0.118	2.045	0.329	3.707

Note: Robust t -statistics are shown in parentheses, clustered at the study level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Four potential outliers are estimated elasticity > 1 .

The regression does not converge when only calculating the estimates without controlling the variable of agglomeration.

Table 4. Moderator variables for meta-regression analysis (observations = 139).

Dimension	Moderator variable		Definition	Mean (SD)
Explained variable	<i>Poly_result</i>		1 = if the findings of polycentricity in the primary study are significantly negative 2 = if the findings of polycentricity in the primary study are non-significant (base) 3 = if the findings of polycentricity in the primary study are significantly positive	1.856 (0.873)
Three commonly criticized specification characteristics (TCCSC)	Measurement of polycentricity	<i>Meas_morph</i>	= 1 if the polycentricity is measured in a morphological way	0.870 (0.336)
		<i>Dum_method</i>	= 1 if the polycentricity is measured based on proportion	2.007 (0.785)
			= 2 if the polycentricity is measured based on rank size (base)	
			= 3 if the polycentricity is measured based on standard deviation	
			= 4 if the polycentricity is measured based on other method	
		<i>Numb_All</i>	= 1 if the polycentricity is measured based on all number of centres	1.331 (0.472)
			= 0 if the polycentricity is measured based on fixed number of centres	
		<i>Dum_scale</i>	= 1 if the primary study is conducted at the micro-level in Asia	4.633 (2.329)
			= 2 if the primary study is conducted at the micro-level in Europe	
			= 3 if the primary study is conducted at the micro-level in North America (no sample)	
			= 4 if the primary study is conducted at the meso-level in Asia	
			= 5 if the primary study is conducted at the meso-level in Europe	
			= 6 if the primary study is conducted at the meso-level in North America	
			= 7 if the primary study is conducted at the macro-level in Asia	
			= 8 if the primary study is conducted at the macro-level in Europe (base)	
			= 9 if the primary study is conducted at the macro-level in North America (no sample)	
	Control variable	<i>Control_agg</i>	= 1 if the study controlled a proxy for agglomeration economies (population or density); 0 otherwise	0.806 (0.397)
	Estimation strategies	<i>Endo_treated</i>	= 1 if the estimate addresses potential endogeneity; 0 otherwise	0.360 (0.482)

(Continued)

Table 4. Continued.

Dimension	Moderator variable	Definition	Mean (SD)
Other study-invariant characteristics (OSIC)	<i>Control_y</i>	= 1 if the explained variable in primary studies is GDP per capita or per labour; 0 otherwise	0.799 (0.402)
	<i>Panel</i>	= 1 if estimate relates to panel data; 0 otherwise	0.280 (0.451)
	<i>Time</i>	Average year of used data	2006.151 (4.135)
	<i>Num</i>	Sample size of primary studies	372.381 (604.634)
Publication-related characteristics (PRC)	<i>Published</i>	= 1 if the estimate comes from a study in an academic journal	0.726 (0.447)
	<i>Pub_year</i>	Year in which the primary study is published	2015.993 (2.793)
	<i>Dum_Eng</i>	= 1 if the study is written in English	0.580 (0.496)

- Since we adopt a general sense of ‘economic performance’, introducing a dummy variable (*Control_y*) to differentiate the GDP per capita/labour-dominated explanatory variable from others is necessary.
- We divide the initial estimates into three categories (*Dum_region*): Asia, Europe and North America.
- The characteristics of data, especially whether they are panel or cross-section data, directly determine the source of variation. We therefore capture this information with a dummy variable (*Panel*).
- The year the data (*Time*) helps to trace the evolution of the effect size, emphasizing trends and possible structural breaks in the literature.
- Sample size (*Num*) may play a vital role in every empirical result.

The third category of variables relates mainly to the publication characteristics (*PRC*) of the primary studies.

- We differentiate published studies with a dummy variable (*Published*).
- We code the information of the year when the studies are published (*Pub_year*).
- A dummy variable is introduced to identify studies written in English (*Dum_Eng*). Table 4 provides the integrated list of variables used in our meta-analysis.

Estimation results

Table 5 reports the estimation results of the meta-logit regression analysis with different specifications. While columns (1–3) separately consider the three different sets of study characteristics, column (4) considers all three sets of moderator variables. Column (5) conducts the backward stepwise regression to filter out the critical variables, and on top of that column (6) performs the meta-logit regression with only English samples (or the likelihood maximization algorithm fails to converge).

In terms of the three oft-criticized specification differences, we do believe they play an important role in explaining the polycentricity–productivity effect. As shown in

Table 5, the pseudo- R^2 for the three commonly criticized specification differences can explain approximately 40% of the variation. In contrast, other study-invariant dimensions, or the standard set of moderators that are commonly controlled for in the meta-regression analysis, account for 20% of the conflicting results. Furthermore, the publication-related characteristics constitute only 12% of the inconclusive findings.

Specifically, the measurement methods of polycentricity and whether agglomeration economies are taken as control variables affect the polycentricity–productivity effect. Compared with rank size-based polycentricity, the standard deviation-based indicators (*Method_std_dev*) are more likely to be significant, with both positive and negative findings. Polycentricity in Asia both at the micro- and meso-levels also has a slight tendency to influence the polycentricity–productivity, which is confirmed by the coefficient of *Meso_Asian* and by the stepwise regression keeping the variable of *Micro_Asian*. Moreover, when considering the agglomeration economies (*Control_agg*), we find a rather robust significant polycentricity–productivity effect as well.

However, the measure of polycentricity from different perspectives (*Meas_morph*) and whether potential endogeneity is addressed (*Endo_treated*) do not appear to have noticeable effects on the estimated economic performance. The insignificance of *Meas_morph* might be partly due to 87% of the extracted estimations in our study being based on morphological polycentricity. Further research with functional polycentricity is greatly needed. The insignificance of *Endo_treated* is predictable because the reconstruction of spatial patterns is a long-term process characterized by high path dependence (Lee, 2006; Lee & Gordon, 2007). In this sense, the spatial structure can be treated as exogenous; hence, recursive causality is not as severe as expected.

Regarding other moderators, we find that studies with panel or the most recent data (*Panel*, *Time*) and published earlier (*Pub_year*) have a higher probability of finding a significantly result. While the odds of finding a significantly negative estimate are higher for studies with a large size (*Num*), the odds of finding a significantly

Table 5. Estimates of meta-analysis with the polynomial logit model.

	(1) TCCSC	(2) OSIC	(3) PRC	(4) All	(5) Stepwise	(6) English
<i>Panel A1: The findings of polycentricity–productivity effect in primary studies are significantly negative</i>						
<i>Meas_morph</i>	–1.666 (–1.62)			2.555 (0.75)		
<i>Method_prop</i>	2.673** (2.57)			4.391 (1.53)		
<i>Method_std. dev.</i>	0.659 (0.52)			8.987 (0.24)	7.632*** (3.73)	23.93 (0.00)
<i>Method_other</i>	–1.970 (–1.30)			–3.406* (–1.71)	–3.900** (–2.47)	–6.015*** (–2.69)
<i>Numb_All</i>	–2.344** (–2.47)			–4.028 (–1.47)		
<i>Micro_Asian</i>	16.16 (0.02)			11.46 (0.01)	7.085 (0.00)	–5.005 (–0.00)
<i>Micro_Euro</i>	16.38 (0.00)			20.04 (0.00)		
<i>Meso_Asian</i>	2.443*** (2.83)			5.401* (1.84)		
<i>Meso_Euro</i>	–0.795 (–0.89)			0.801 (0.18)		
<i>Meso_N-America</i>	–17.13 (–0.01)			–15.91 (–0.00)		
<i>Control_agg</i>	1.546* (1.69)			6.977** (2.41)	8.473*** (4.60)	28.79 (0.01)
<i>Endo_treated</i>	0.368 (0.49)			–1.552 (–1.19)		
<i>Control_y</i>		1.715*** (2.64)		1.942 (0.51)		
<i>Panel</i>		1.024 (1.50)		9.598 (0.26)	8.099*** (4.08)	–
<i>Time</i>		0.223*** (2.86)		1.450*** (3.15)	1.592*** (4.37)	2.668*** (3.66)
<i>Num</i>		0.005*** (3.25)		0.029*** (2.97)	0.0364*** (4.65)	0.0509*** (4.07)
<i>Published</i>			0.0954 (0.16)	–6.426** (–2.32)	–5.152*** (–3.59)	–11.00 (–1.64)
<i>Pub_year</i>			0.282*** (3.19)	–1.430*** (–2.79)	–1.380*** (–3.81)	–2.021*** (–3.49)
<i>Dum_Eng</i>			–0.165 (–0.31)	0.759 (0.02)		
<i>Panel A2: The findings of polycentricity–productivity effect in primary studies are significantly positive</i>						
<i>Meas_morph</i>	1.656 (1.13)			2.917 (0.98)		
<i>Method_prop</i>	–0.228 (–0.20)			1.383 (0.52)		
<i>Method_std. dev.</i>	0.915 (0.74)			4.932 (0.13)	3.000* (1.83)	3.285 (0.00)

(Continued)

Table 5. Continued.

	(1) TCCSC	(2) OSIC	(3) PRC	(4) All	(5) Stepwise	(6) English
<i>Method_other</i>	−2.824* (−1.73)			0.891 (0.20)	−1.974 (−1.09)	−23.28 (−0.00)
<i>Numb_All</i>	−1.449 (−1.64)			−3.232 (−1.18)		
<i>Micro_Asian</i>	16.82 (0.02)			20.57 (0.01)	16.91 (0.01)	19.72 (0.00)
<i>Micro_Euro</i>	1.013 (0.00)			10.31 (0.00)		
<i>Meso_Asian</i>	0.583 (0.67)			5.232* (1.86)		
<i>Meso_Euro</i>	−17.37 (−0.01)			−15.09 (−0.00)		
<i>Meso_N-America</i>	−0.688 (−0.63)			12.33 (1.61)		
<i>Control_agg</i>	1.506 (1.41)			1.244 (0.47)	2.536** (2.13)	27.27 (0.00)
<i>Endo_treated</i>	0.419 (0.56)			−0.416 (−0.37)		
<i>Control_y</i>		2.121*** (3.05)		0.653 (0.45)		
<i>Panel</i>		1.321** (2.00)		2.974 (0.08)	3.874*** (3.12)	—
<i>Time</i>		0.193*** (2.66)		0.200 (0.76)	0.237* (1.66)	1.335** (2.19)
<i>Num</i>		0.001 (0.77)		−0.001 (−0.09)	0.003 (0.98)	0.013 (1.45)
<i>Published</i>			1.473** (2.27)	−1.292 (−0.51)	0.431 (0.42)	−6.369 (−1.03)
<i>Pub_year</i>			0.137 (1.39)	0.856 (1.02)	−0.208 (−1.18)	−0.871** (−2.10)
<i>Dum_Eng</i>			−1.723*** (−3.01)	−5.024 (−0.13)		
Observations	139	139	139	139	139	80
Number of studies	31	31	31	31	31	17
Pseudo- R^2	0.40	0.20	0.12	0.70	0.58	0.71
Log-likelihood	−87.47	−116.95	−129.31	−43.41	−61.77	−23.69

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

t -statistics are shown in parentheses.

Moderator variables of *Micro_N-America*, *Meso_Asia* and *Panel* are omitted because of collinearity.

positive estimate are higher for studies published earlier (*Pub_year*).

CONCLUSIONS

Currently, there is a growing number of studies that have evaluated the effect of polycentric spatial structure on economic performance. However, the findings of the polycentric-productivity effect have been quite inconclusive, and little work on this topic has generalized across studies

or helped to understand the differing results. Worse, this inconclusiveness can cause substantial trouble because the conception of polycentricity is closely related to urban planning practice. Therefore, this paper attempts to systematically review and analyse previous studies by means of a meta-analysis and to clarify the relationship between polycentricity and economic performance.

As the first meta-analysis in this field, we sample 139 estimates from 31 primary studies. After analysing the vote count, funnel plot and precision-effect test, we find

no compelling evidence to support that polycentricity is either beneficial or detrimental for economic development. Additionally, we find no evidence of publication selection bias. Further meta-logit regressions confirm that the influences of the three oft-criticized specification differences do play an important role. Specifically, the measurement methods of polycentricity and adopting agglomeration economies as a control variable affect the polycentricity–productivity effect. In contrast, the measure of polycentricity from different perspectives and whether potential endogeneity is addressed do not have noticeable effects on the estimated economic performance.

Our finding of no evidence to support the idea that polycentricity is either favourable or harmful to economic development calls for critical reflection. In fact, previous studies have already questioned the positive effects of polycentricity, which are largely based on individual empirical research or observations that lack a theoretical basis (Brezzi & Veneri, 2015; Davoudi, 2003; Meijers & Sandberg, 2008). In this paper, we confirm that there is no solid evidence for polycentricity promoting economic development. However, this paper does not generally invalidate the utility of polycentricity. First, our finding is an overall result of the offsetting positive and negative effects of polycentric structure on the economy, and the influence of polycentric structure on economic development is more than likely to be significant in specific cases. Second, there could be other reasons for pursuing polycentric development, such as reducing disparities or promoting the environmental sustainability. Third, this paper is an account of only a handful of empirical studies, and the meta results will be more credible when drawing on considerable empirical studies. Nevertheless, there is a pressing need for more empirical examinations of the relationship between polycentricity and its various performances, especially from the perspective of functional polycentricity.

Our findings also provide helpful guidance for future research in this area. First, achieving a universally accepted definition and identification of polycentricity and thus increasing the comparability of studies remains a worthwhile challenge in this field. According to our meta-logit regression, the measurement method and scale of polycentricity do affect the relationship between polycentricity and economic development, and the insignificance of morphological polycentricity may be due to the limitation of samples. Second, the effect of agglomeration economies on polycentricity–productivity must not be overlooked. The analysis shows that agglomeration economies constitute a critical control variable, and studies are more likely to find a negative polycentricity–productivity effect when considering agglomeration economies. Third, endogeneity bias of the polycentricity–productivity effect may not be a significant issue. Treatment of the endogeneity of polycentricity does not appear to produce any significant difference. Future studies do not need to worry excessively about the issue of reverse causality.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the authors.

FUNDING

This work is supported by the National Natural Science Foundation of China [grant numbers 41901184 and 42071210]; the China Postdoctoral Science Foundation [grant numbers 2019M650080 and 2020T130194]; the Major Program of the National Social Science Foundation of China [grant number 17ZDA068]; the Fundamental Research Funds for the Central Universities [grant number 2019ECNU-HWFW002]; and the Key Scientific Research Program of Higher Education Institutions in Henan Province [grant number 22A170006].

NOTE

1. Five studies did not have keywords, and we extracted these terms from the titles and abstracts of these studies. Specifically, the keywords extracted for Meijers (2008) were polycentric urban region, provision of amenities, monocentricity and Dutch region; the keywords extracted for Meijers and Burger (2010) were spatial structure, productivity, agglomeration, monocentricity, polycentricity and US metropolitan area; the keywords extracted for Veneri and Burgalassi (2012) were polycentric development, functional and morphological polycentricity, economic performance, and Italian NUTS-2 region; the keywords extracted for Brezzi and Veneri (2015) were polycentric urban system, economic implication, polycentricity, spatial scale and OECD countries; and the keywords extracted for Ouwehand (2018) were polycentricity, spatial organization, productivity, polycentric city networks and European TL2 region.

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